

Executive Overview

Executive summary of financial performance and fraud metrics.

2019

2020



Transaction Amount

\$91M



Total Transactions

1M



Fraud Rate %

0.6%



Fraud Loss Amount

\$4.0M



Avg Transaction Value

70.35

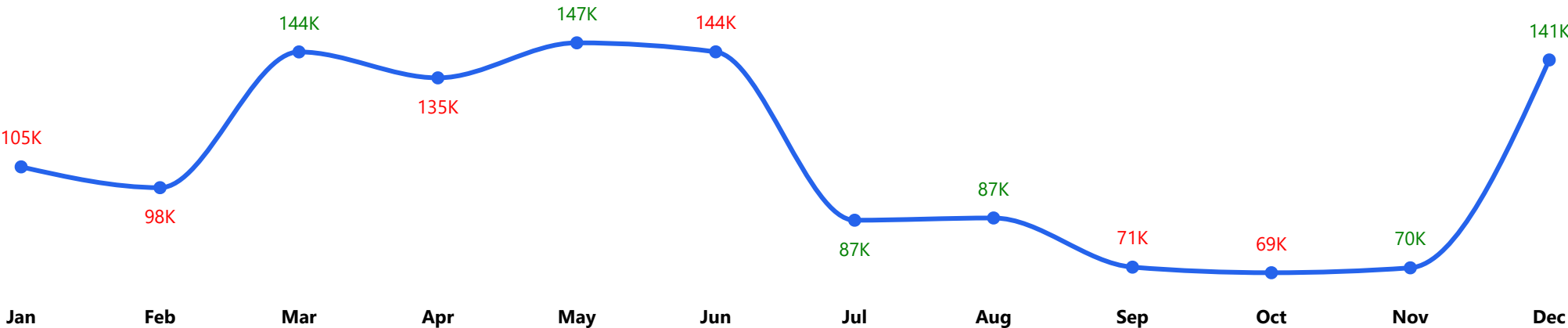
Key Business Insight

grocery_pos accounts for 15.9% of total transaction value.

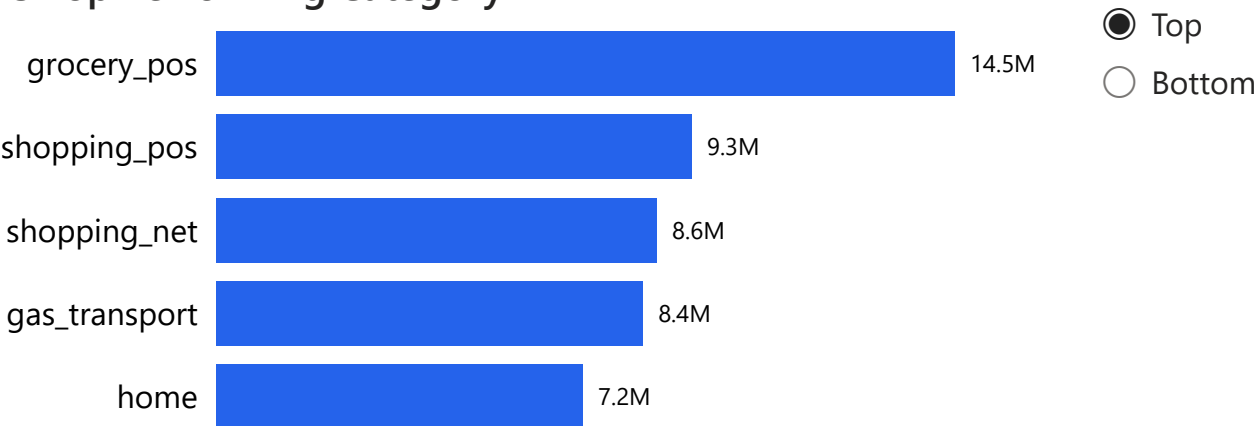
Performance Insight

TX generated the highest transaction value of \$6.8M.

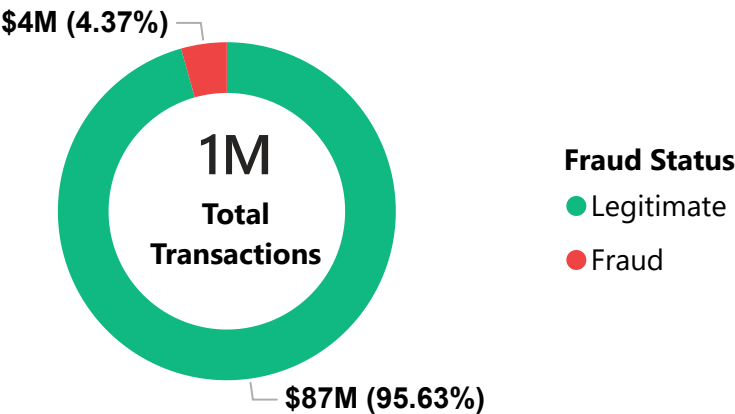
Total Transactions by Month Name



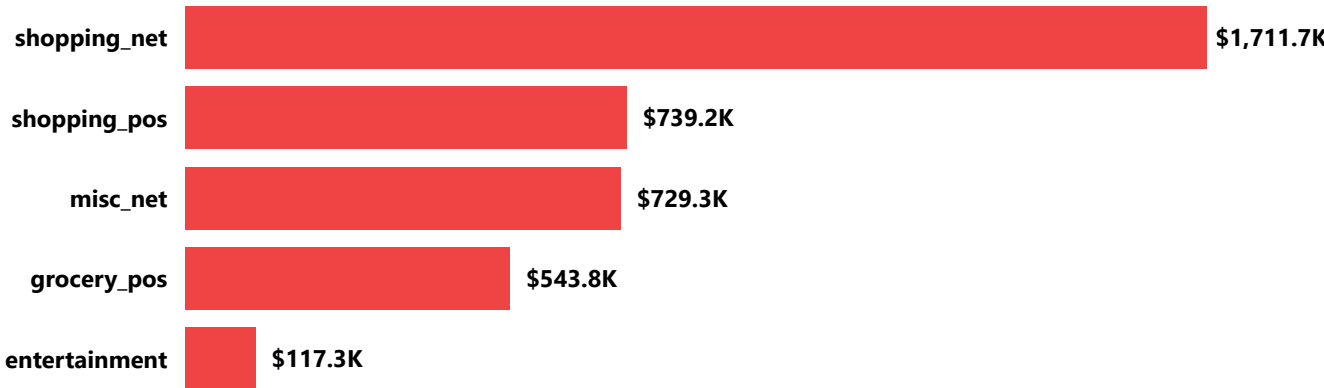
5 Top Performing Category



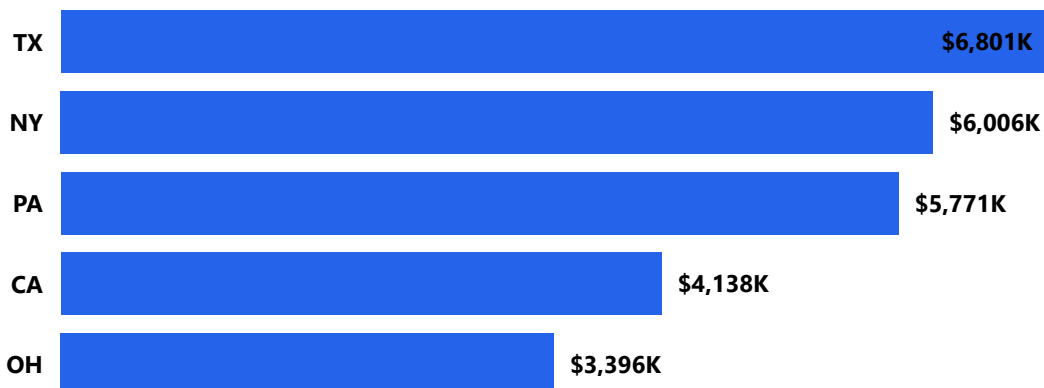
Transaction Amount by Fraud Status



Fraud Loss Amount by category



Transaction Amount by State



FRAUD & RISK ANALYSIS

Analyze fraud patterns, locations and transaction risk.

2019

2020



Fraud Transactions

8K



Fraud Rate %

0.6%



Fraud Loss Amount

\$4.0M



Highest Risk Category

shopping_net



Highest Fraud Rate %

1.8%

Fraud Alert

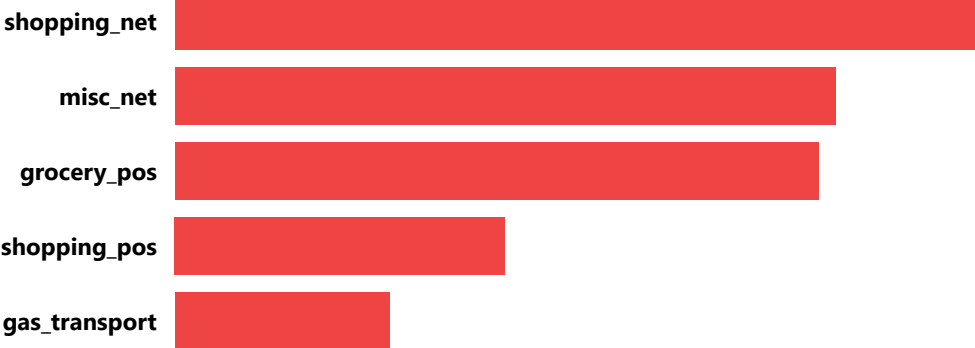
Kozey-Boehm recorded the highest fraud loss of 48.2K

High-risk fraud activity is concentrated between 10:00 PM and 3:00 AM, accounting for 6,362 fraud transactions (84.8% of all fraud cases). This period should be prioritized for enhanced fraud monitoring.

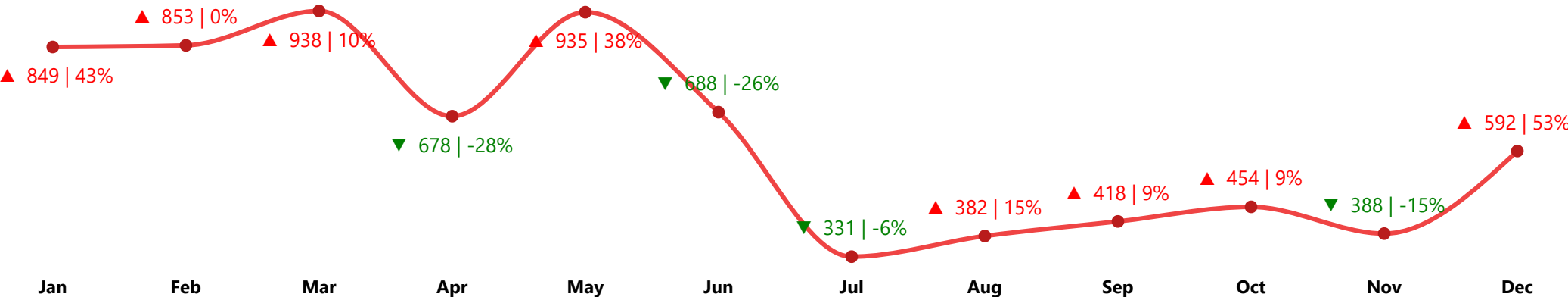
5 Top Fraud Categories

Top

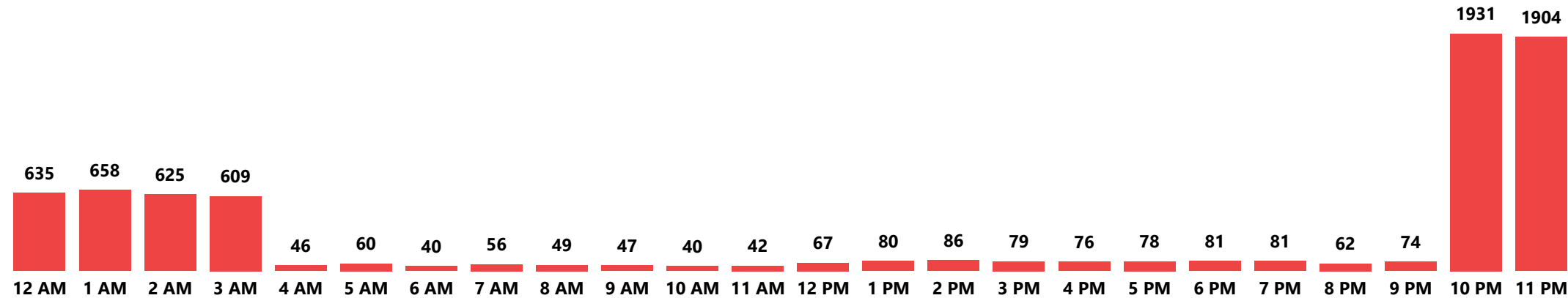
Bottom



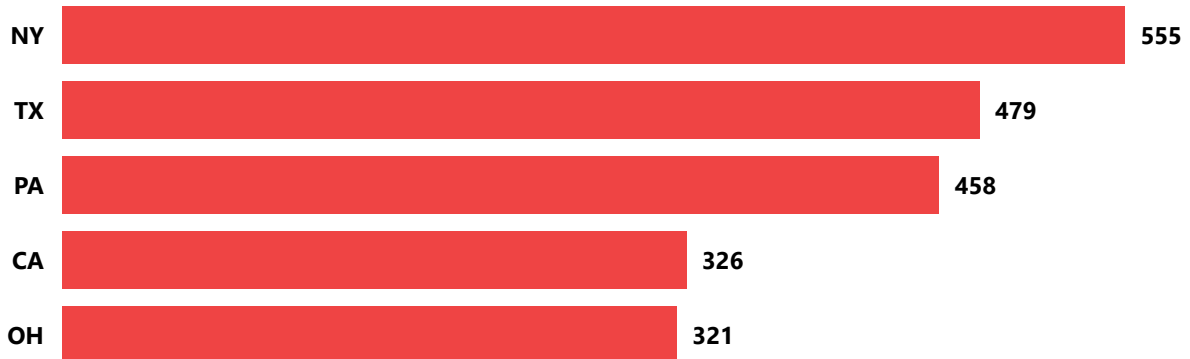
Fraud Transactions by Month



Fraud Transactions by Hour of the day



Fraud Transactions by state



CUSTOMER BEHAVIOR ANALYTICS

Explore customer spending behavior and demographic trends.

2019

2020

Total Customers

983

Avg Spend per customer

92.80K

Highest Spending Age Group

35-44 \$22.3 (24.5%)

Highest Spending Gender

Female \$49.8M (54.6%)

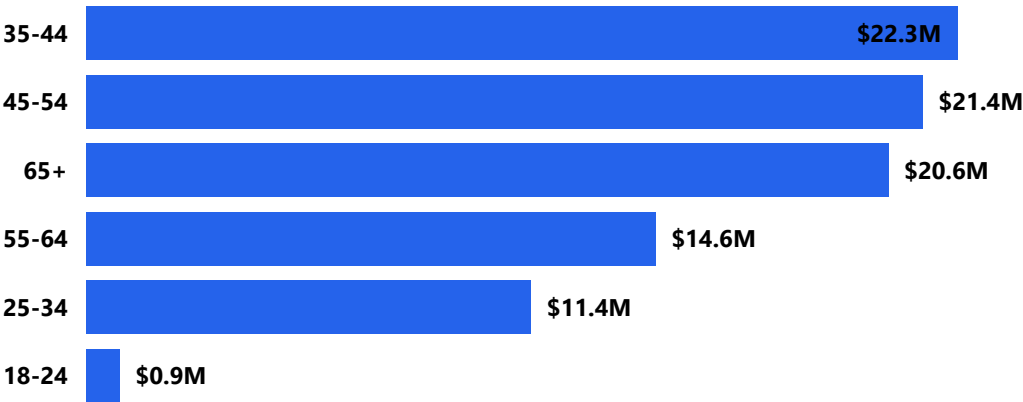
Customer Insight

Customers aged 35-44 contribute 24.5% of total spending.

Spending Pattern

Female customers account for 54.6% of total transaction value.

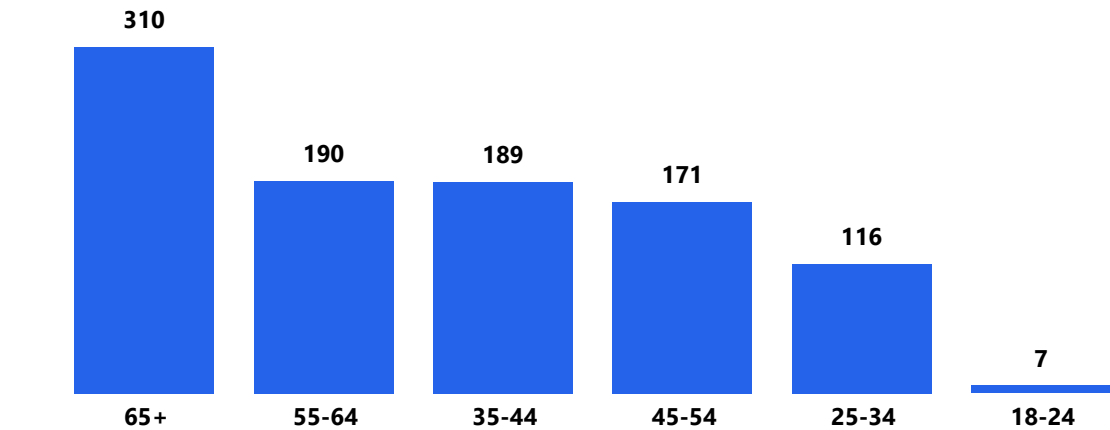
Transaction Amount by Age Group



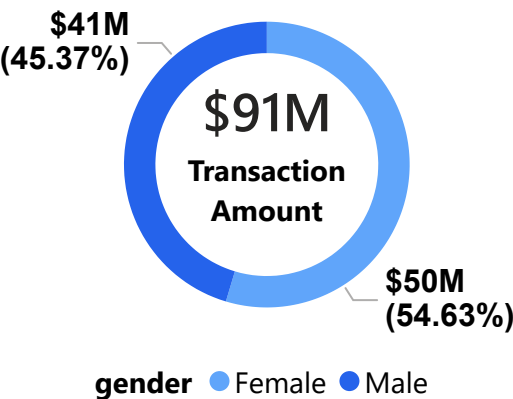
Top 5 Customers by Transaction



Total Customers by Age Group



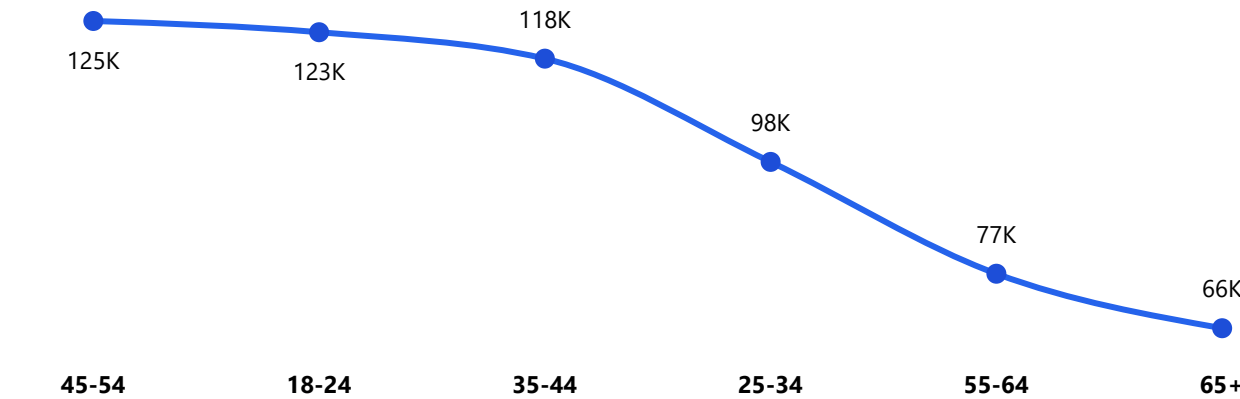
Transaction Amount by gender



Transaction Amount by job



Average Spend by Age Group



MERCHANT AND BUSINESS INSIGHTS

Evaluate merchant performance and fraud exposure.

2019

2020

Active Merchants

693

Top Merchant by Revenue

Kilback LLC

Avg Merchant Revenue

131.63K

Highest Risk Merchant

Kozey-Boehm

Merchant Fraud Rate

0.6%

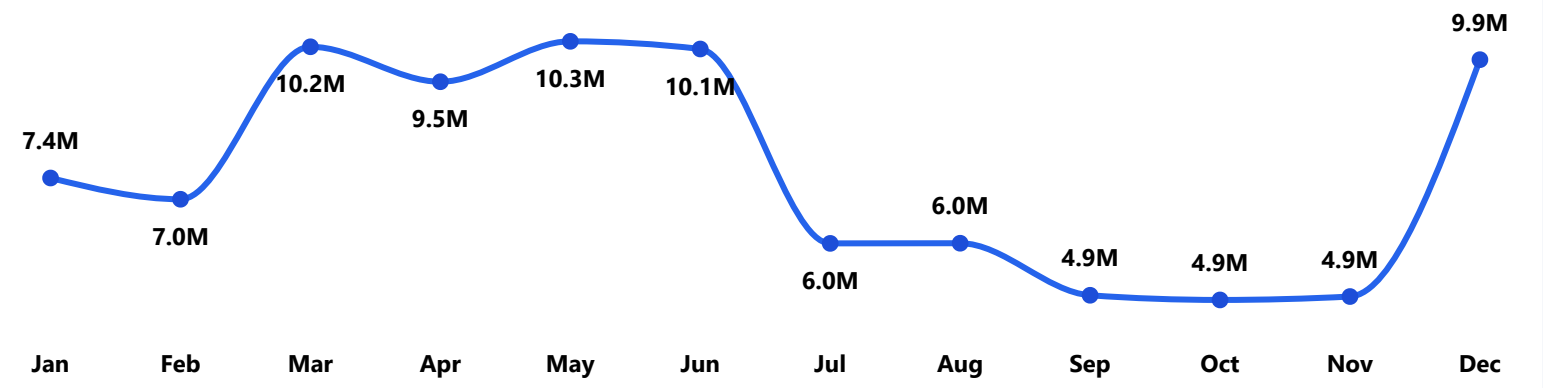
Kilback LLC generated the highest transaction value of \$391.1K, making it the highest-performing merchant in the selected period.

The Top 10 merchants contribute 3.4% of total transaction value, indicating a highly diversified merchant network.

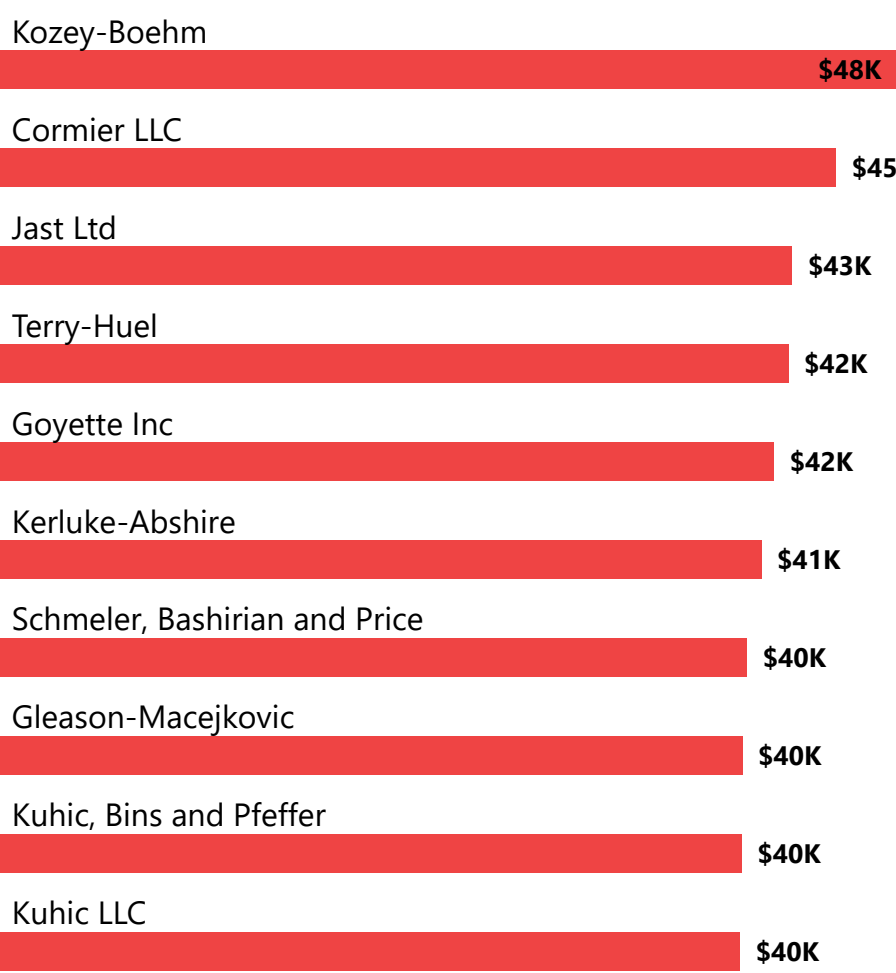
Top 5 merchant by Revenue



Merchant Revenue Monthly Trend



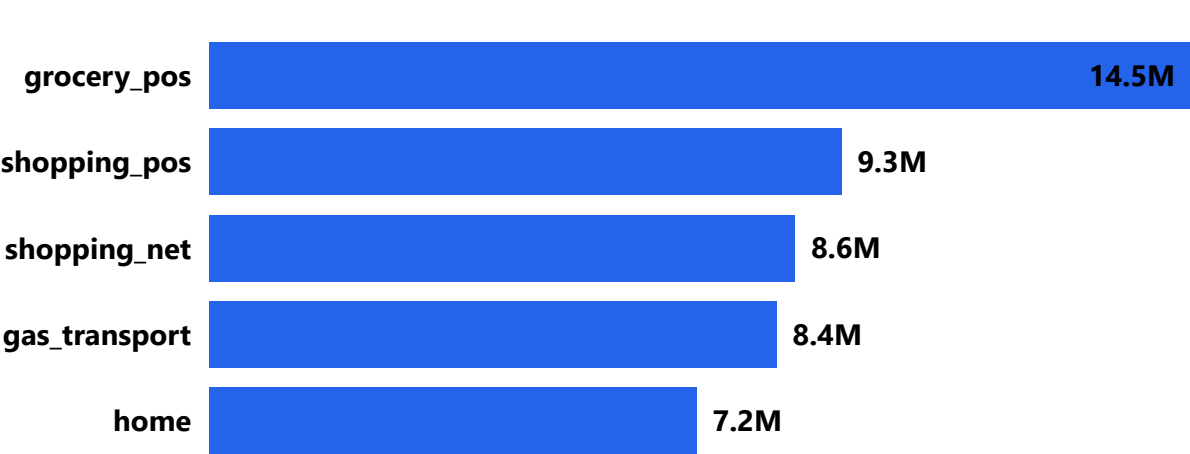
Top 10 merchant Fraud Loss



Fraud Rate by merchant



Merchant Revenue by category



Executive Summary

This dashboard analyzes **925K+ credit card transactions** from **2019 - 2020** to uncover trends in transaction performance, fraud patterns, customer behavior, and merchant performance. The analysis provides actionable insights to support fraud prevention, customer engagement, and business decision making.

Business Questions Answered

- Which categories and states generate the highest transaction value?
- Which merchant categories and merchants have the highest fraud exposure?
- When is fraudulent activity most likely to occur?
- Which customer segments contribute the most revenue?
- Is merchant revenue concentrated or well diversified?

Key Insights

- **Grocery POS** generated the highest transaction value, contributing approximately **16%** of total spending.
- Fraud activity was highest between **10:00 PM and 3:00 AM**, accounting for over **84%** of fraud cases.
- **Customers aged 35 - 44** and **female customers** contributed the highest transaction value.
- The **Top 10 merchants** contributed only **3 - 4%** of total transaction value, indicating a well diversified merchant network.

Recommendations

- Increase fraud monitoring during **10 PM - 3 AM**.
- Apply stricter controls to high risk shopping related categories.
- Closely monitor merchants with elevated fraud rates.
- Strengthen customer retention strategies for the **35 - 44** age segment.